Robotic Systems

Chapter I
Robotics; an Introduction

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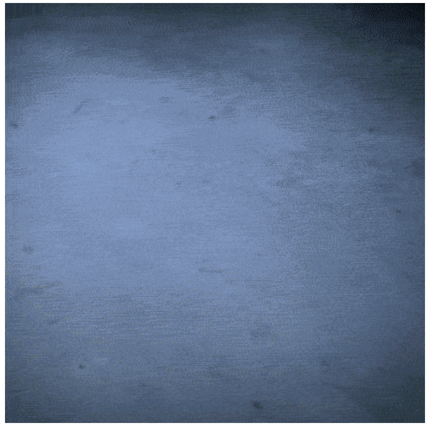
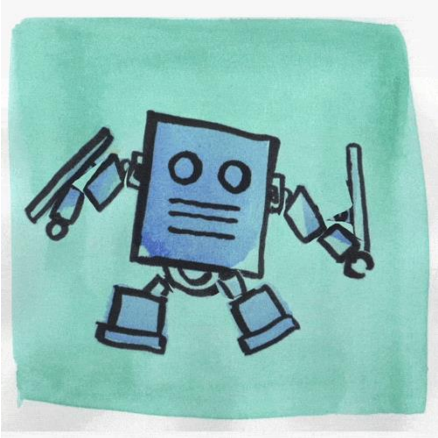
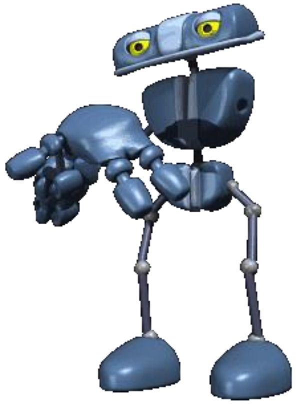
What Is a Robot?

Robots are a diverse bunch. Some walk around on their two, four, six, or more legs, while others can take to the skies. Some robots help physicians to do surgery inside your body; others toil away in dirty factories. There are robots the size of a coin and robots bigger than a car. Some robots can make pancakes. Others can land on Mars.

This diversity—in size, design, capabilities—means it's not easy to come up with a definition of what a robot is.

What is a robot?

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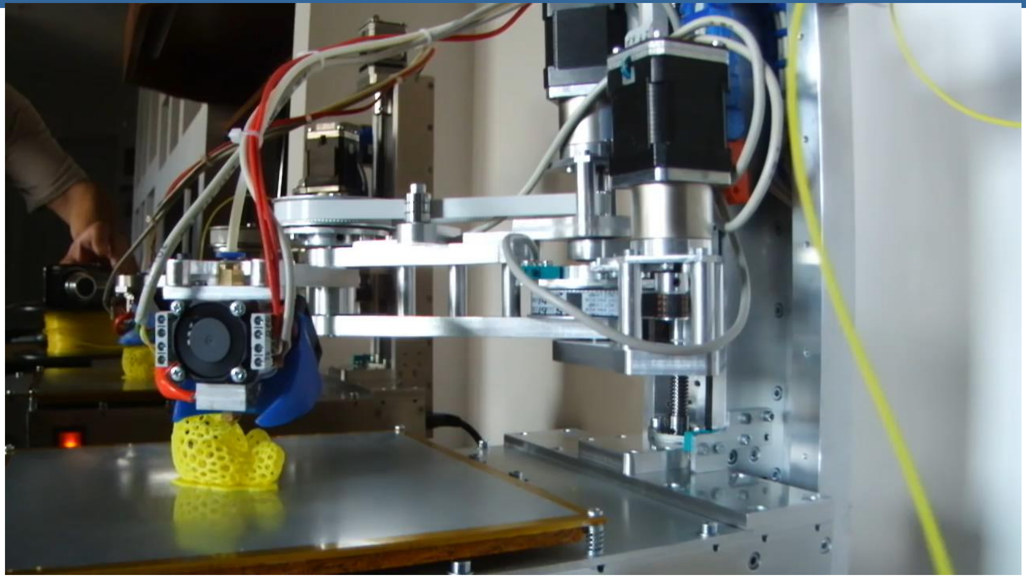
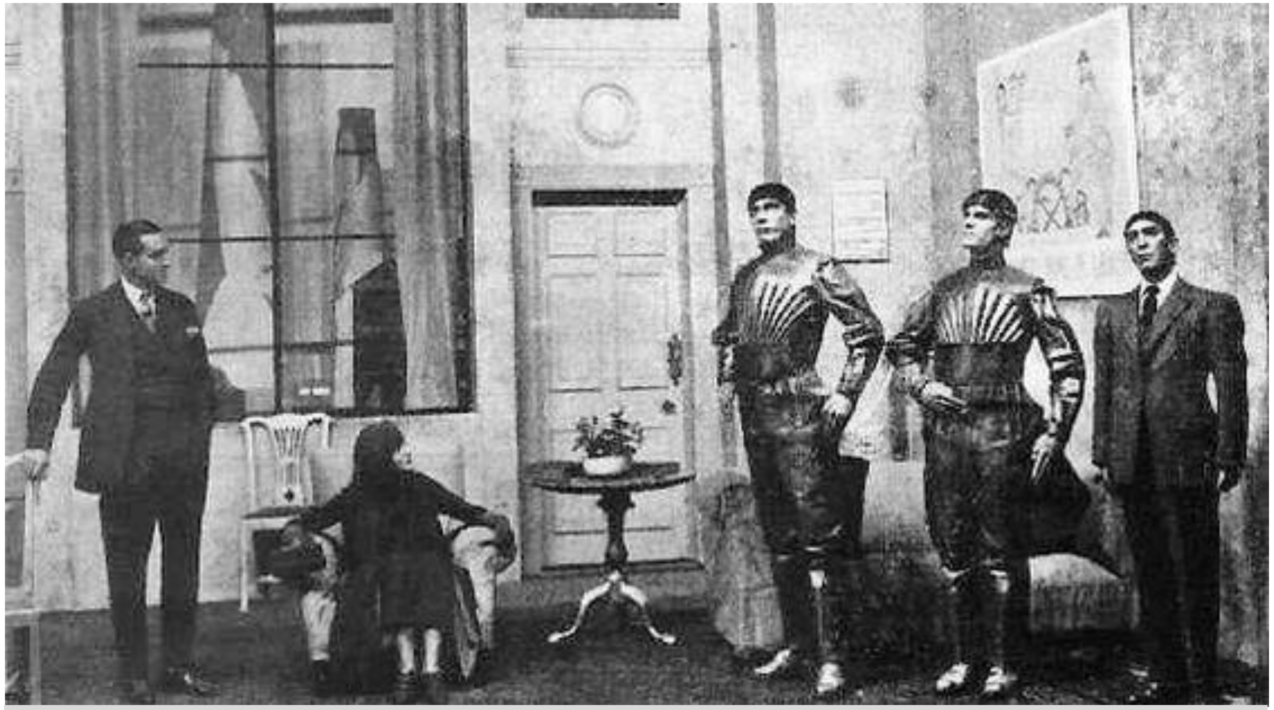


Robot



A scene from the play, showing three robots

Written by	Karel Čapek
Date premiered	25 January 1921
Original language	Czech
Genre	Science fiction

Definition:

"A **robot** is **defined** as a reprogrammable, multifunctional manipulator designed to move material, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks."

Robot Institute of America

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Merriam-Webster SINCE 1828

robot

DICTIONARY THESAURUS

robot noun

ro·bot | \ ˈrō-ˌbät, -bət\

plural **robots**

Definition of *robot*

1 : a machine that resembles a living creature in being capable of moving independently (as by walking or rolling on wheels) and performing complex actions (such as grasping and moving objects)

A robot is an autonomous machine capable of sensing its environment, carrying out computations to make decisions, and performing actions in the real world.

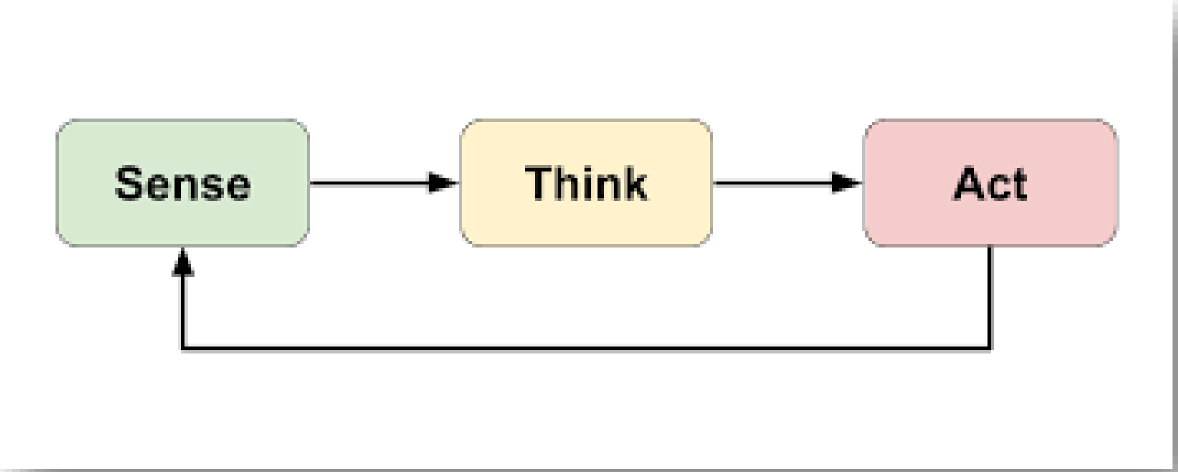
Gill Prat, Toyota Research Institute

What is a robot?

- A device that has sensors to sense the outside world
- Has some sort of computation
- Decides an action according to the sensors
- Do an action or change outside it's body

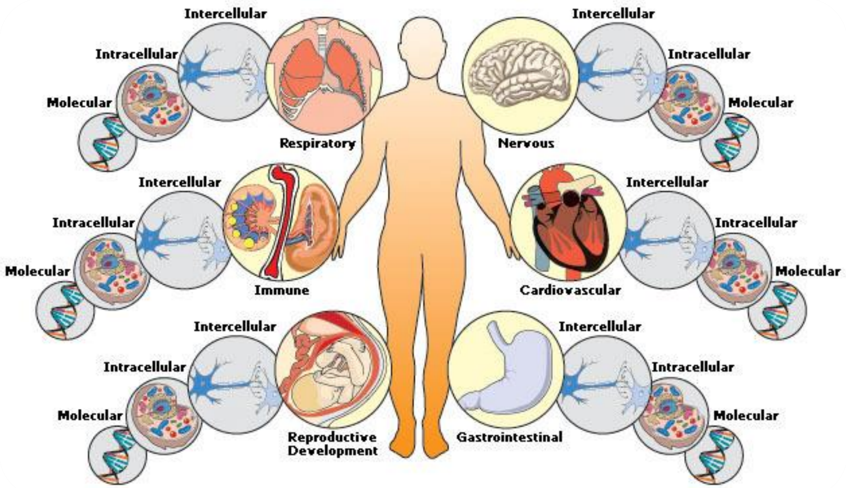
also

- What it does and how it does?



A **robot** is a system.

System: a group of related elements that work together to perform a specific job.



Robot Types

1. **By Application**
2. **By locomotion and Kinematics**

1. Robots do a lot of different tasks in many fields and the number of jobs entrusted to robots is growing steadily. That's why one of the best ways how to divide robots into types is a division by their application.

Industrial robots

Industrial robots are robots used in an industrial manufacturing environment. Usually these are articulated arms specifically developed for such applications as welding, material handling, painting and others.

Domestic or household robots

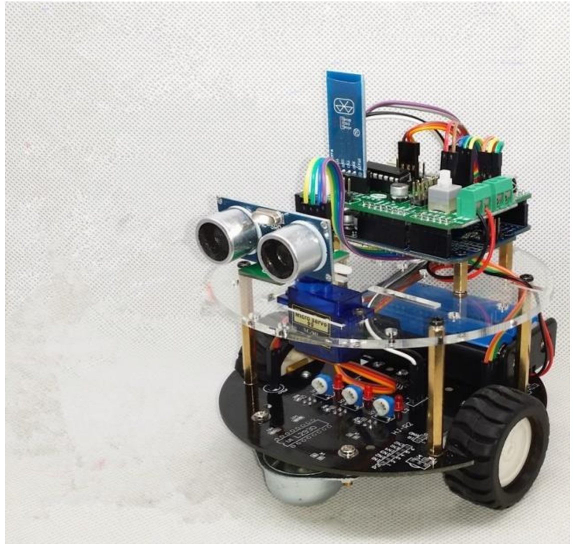
Robots used at home. This type of robots includes many quite different devices such as robotic vacuum cleaners, robotic pool cleaners, sweepers, gutter cleaners and other robots that can do different chores. Also, some surveillance and telepresence robots could be regarded as household robots if used in that environment.

Medical robots

Robots used in medicine and medical institutions. First and foremost - surgery robots. Also, some automated guided vehicles and maybe lifting aides.

Service robots

Robots that don't fall into other types by usage. These could be different data gathering robots, robots made to show off technologies, robots used for research, etc.



Military robots

Robots used in military. This type of robots includes bomb disposal robots, different transportation robots, reconnaissance drones. Often robots initially created for military purposes can be used in law enforcement, search and rescue and other related fields.

Entertainment robots

These are robots used for entertainment. This is a very broad category. It starts with toy robots such as Robosapiens or the running alarm clock and ends with real heavyweights such as articulated robot arms used as motion simulators.

Space robots

This type would include robots used on the International Space Station, Canadarm that was used in Shuttles, as well as Mars rovers and other robots used in space.

2. Types of robots by locomotion and kinematics

Robot's application alone does not provide enough information when talking about a specific robot. For example ,when talking about industrial robots we think of stationary robots in a work cell that do a specific task. That's alright, but if there is an AGV (Automated Guided Vehicle) in a factory? It's also a robotic device working in an industrial environment. So, to describe the robot in the best way, it is necessary to use both these classifications together.

Stationary Robots

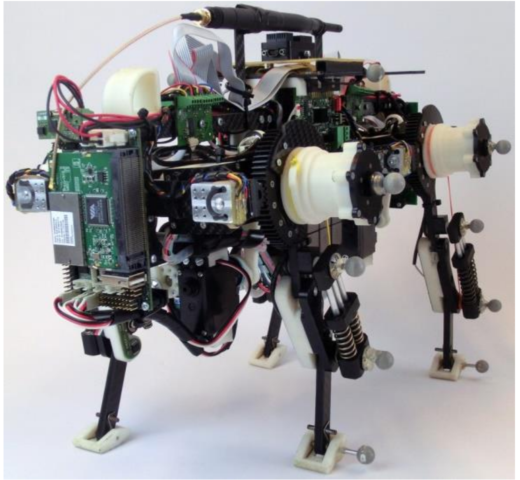
- Cartesian
- Cylindrical
- Spherical
- Flying

Wheeled Robot

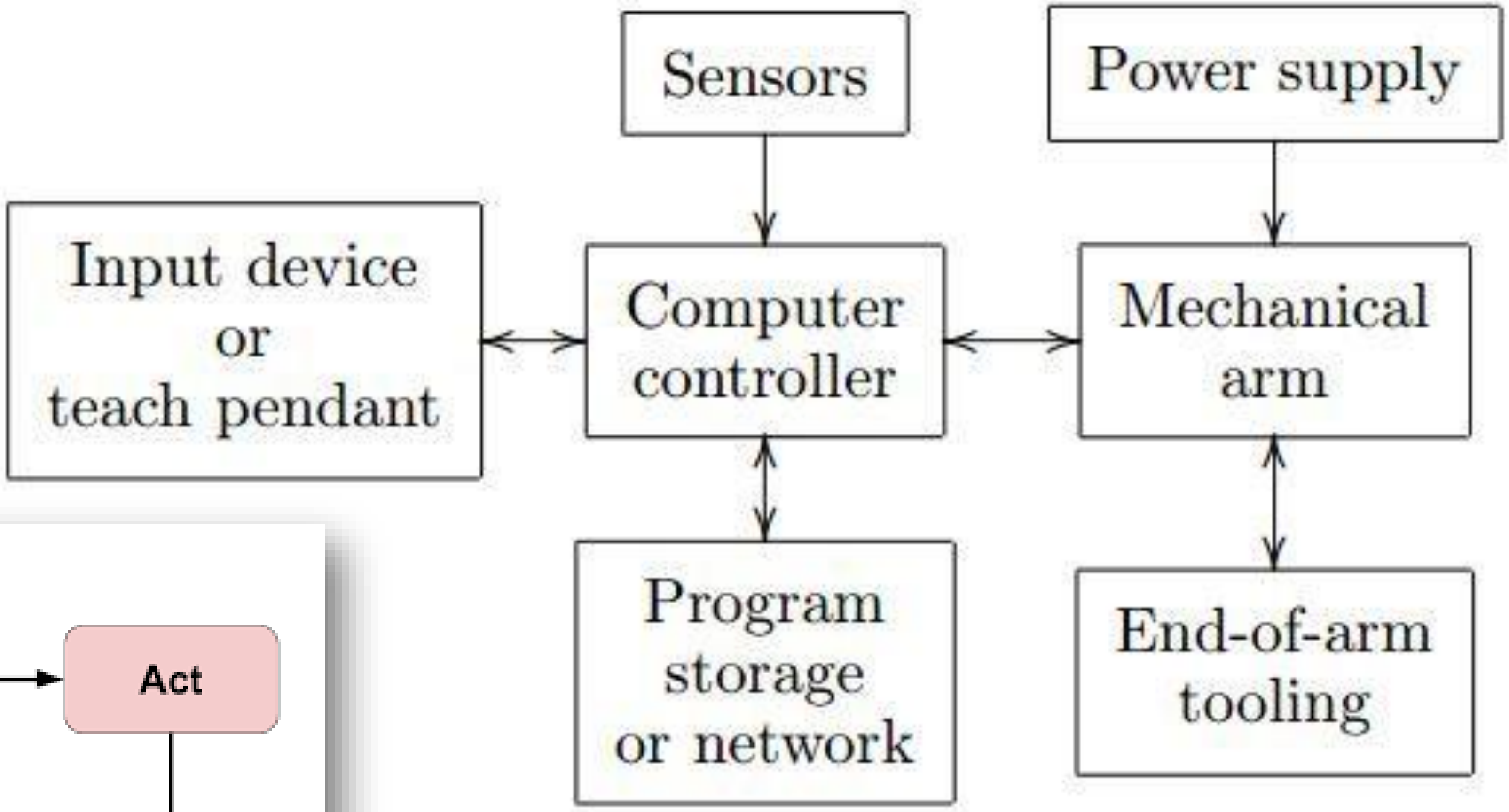
Single, two, more wheels

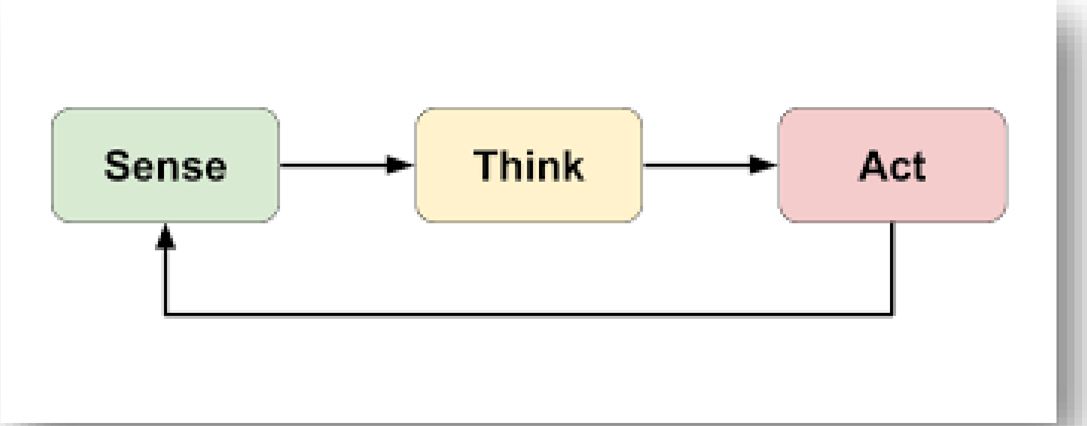
Legged robots

Bipedal robots(humanoid robots)
Tripedal robots
quadrupedal robots
hexapod robots



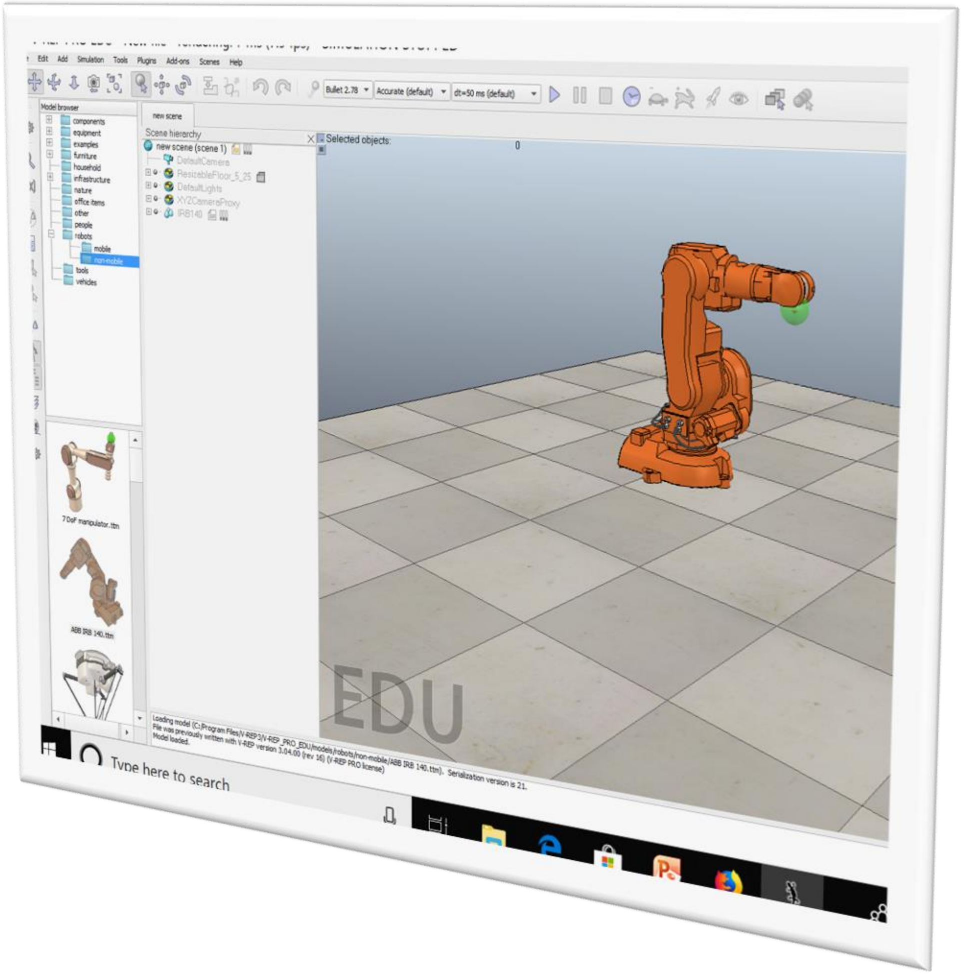
Robotic Systems Architecture





Robotic Platforms

MATLAB
MATLAB ROBOTIC TOOLBOX
V-REP
GAZEBO
ROBOLOGIX
ROS
Jasmine
…….



Robotic Module (CE513)

- This is an introductory course in robotics for postgraduate students who have had little to no other introductory courses on the topic.

- The course focuses on topics in robotics that relate to architecture, modeling, analysis, dynamics, and control of robotic manipulators.

- The course covers robot configuration, forward and inverse Kinematics, mathematical modelling of robots with an emphasis on planning algorithms, fundamentals of robot sensors, sensor processing algorithms, robot control and programming.

Resources

- Kevin M. Lynch and Frank C. Park. 2017. Modern Robotics, Cambridge University Press.
- Damith, H., and David S.,2022. Foundations of Robotics, Springer.
- Mztjaz Mihelij, et. al, 2019. Robotics, Springer.

Robot Labs

https://www.skyfilabs.com

https://robotacademy.net.au/

Assessments					
	No.	Time	Weight (%)	Week Due	Notes
Team Project	1		30	9-14	Class Presentation
Mid-Term Exam	1	2 Hour	20	8	Scheduled
Exam (Final)	1	2 hours	50	15	Scheduled

Thank you